

DZHARDZHAN, Russian Federation

WMO: 241430

Lat: **68.735N**Lon: **124.021E**Elev: **126**StdP: **14.63**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-57.5	-54.4	-61.3	0.1	-56.8	-58.1	0.2	-54.1	38.2	-15.4	31.6	-14.1	1.3	20	0.858

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	78.3	64.9	74.4	62.6	70.5	60.4	67.0	75.2	64.5	71.9	61.8	68.9	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	88.9	71.5	61.3	81.4	69.3	58.4	73.1	66.1	31.7	75.8	29.8	72.0	27.8	69.2	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
30.6	25.8	20.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-62.3	83.9	2.9	3.3	-64.4	86.3	-66.1	88.3	-67.8	90.1	-69.9	92.5		
			-62.2	70.9	2.8	2.6	-64.3	72.8	-65.9	74.4	-67.5	75.9	-69.5	77.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-33.8	-30.7	-8.4	16.4	33.8	53.3	58.9	53.5	38.7	13.4	-19.5	-33.8
	DBStd	36.29	12.72	13.24	15.04	11.78	10.62	8.85	7.78	6.19	6.81	12.96	13.84	13.02
	HDD50	14451	2598	2258	1809	1009	511	65	9	32	344	1133	2085	2598
	HDD65	19375	3063	2678	2274	1459	968	363	224	361	789	1598	2535	3063
	CDD50	600	0	0	0	0	8	163	285	139	5	0	0	0
	CDD65	49	0	0	0	0	0	11	35	3	0	0	0	0
	CDH74	389	0	0	0	0	4	94	266	25	0	0	0	0
	CDH80	68	0	0	0	0	0	15	53	0	0	0	0	0
Wind	WSAvg	6.8	6.4	5.2	5.9	7.4	7.8	7.7	7.7	6.9	7.4	7.8	5.3	5.7
Precipitation	PrecAvg	12.0	0.5	0.4	0.5	0.5	0.8	1.6	2.0	1.8	1.5	1.5	0.7	0.6
	PrecMax	16.9	1.7	1.2	1.4	1.7	2.4	3.6	5.4	5.3	3.4	3.2	2.0	1.6
	PrecMin	7.4	0.1	0.1	0.1	0.0	0.1	0.1	0.4	0.1	0.3	0.2	0.2	0.1
	PrecStd	2.8	0.3	0.3	0.4	0.4	0.6	0.9	1.2	1.2	0.7	0.7	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	7.6	30.6	47.3	67.5	82.4	85.3	77.4	64.6	35.7	19.5	1.6
		MCWB	1.2	6.4	26.9	38.1	51.3	65.6	68.0	63.4	56.1	34.1	18.0	0.8
	2%	DB	-6.8	0.4	24.7	41.1	59.8	76.5	80.6	72.5	57.2	32.9	14.0	-3.1
		MCWB	-7.6	-0.4	22.2	34.4	47.2	61.4	66.7	61.5	50.2	31.5	13.1	-3.9
	5%	DB	-12.4	-5.3	19.7	37.6	54.7	72.4	76.4	68.8	52.6	30.7	8.5	-8.5
		MCWB	-12.9	-5.9	17.8	32.6	45.0	59.1	65.2	59.4	47.3	29.4	7.5	-8.9
	10%	DB	-16.0	-11.6	14.9	34.4	49.6	68.1	72.4	64.9	49.1	28.4	0.3	-13.8
		MCWB	-16.4	-12.1	13.2	30.3	41.6	57.3	63.3	57.6	44.5	27.1	-0.3	-14.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	6.5	27.1	38.4	51.9	67.9	71.0	66.5	56.5	34.6	18.4	0.8
		MCDB	1.9	7.7	30.4	46.4	65.7	76.8	80.8	73.9	63.8	35.2	19.5	1.3
	2%	WB	-7.4	-0.3	22.6	35.5	48.2	63.7	68.5	63.0	51.6	31.8	13.2	-3.7
		MCDB	-6.7	0.5	24.7	40.0	58.3	73.3	77.4	69.1	55.9	32.5	13.9	-3.1
	5%	WB	-12.8	-6.0	17.8	33.1	45.4	61.0	66.4	60.3	48.0	29.5	7.7	-8.9
		MCDB	-12.4	-5.4	19.7	37.3	53.7	69.0	74.3	66.8	51.7	30.7	8.6	-8.6
	10%	WB	-16.4	-12.0	13.3	30.1	42.2	58.2	64.2	58.2	45.0	27.2	-0.3	-14.2
		MCDB	-16.0	-11.5	14.9	34.6	49.0	66.7	71.2	63.9	48.3	28.3	0.3	-13.8
Mean Daily Temperature Range	5% DB	MDBR	10.7	13.0	18.8	19.8	15.2	16.3	15.6	15.4	12.4	10.2	11.3	10.3
		MCDBR	13.6	16.5	16.1	18.3	21.0	22.8	21.7	22.8	20.2	7.8	13.4	14.4
		MCWBR	13.2	15.8	14.5	13.6	12.3	11.8	11.3	13.1	13.3	7.3	13.3	14.0
	5% WB	MCDBR	13.4	16.6	15.9	16.1	20.0	19.7	19.0	20.0	18.3	7.6	13.4	14.5
		MCWBR	13.1	15.9	14.4	12.6	12.1	12.2	11.0	12.3	13.2	7.3	13.3	14.1
Clear-Sky Solar Irradiance	taub	0.185	0.225	0.284	0.319	0.353	0.363	0.444	0.422	0.308	0.238	0.181	N/A	
	taud	1.873	2.022	2.060	2.081	2.205	2.418	2.155	2.171	2.472	2.121	1.888	N/A	
	Ebn at Noon	18	174	229	257	262	265	232	221	233	165	15	0	
	Edh at Noon	2	15	27	36	36	31	39	34	19	13	2	0	
All-Sky Solar Radiation	RadAvg	8	175	708	1425	1825	1823	1640	1088	597	189	29	0	
	RadStd	1	13	45	64	148	190	134	132	58	28	3	0	